EC7207: High Performance Computing

Analysis Report

Parallel Pearson Correlation Based Recommendation System

Group Number: 11

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Technologies: Serial · OpenMP · Pthreads · MPI · Hybrid MPI+OpenMP

Problem: 1,000 users × 1,000 items · SEED=42 · Sparsity=70% · TOP-K=20

Platform: Linux · GCC 15.2 · OpenMPI · 32 logical CPU cores

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1. Introduction

Modern collaborative filtering recommendation systems compute pairwise user similarity over large rating matrices. As the number of users grows, the $O(N^2 \times M)$ similarity computation becomes the dominant bottleneck — for N = M = 1,000 this equates to approximately 500 million floating-point operations.

This report documents the design, implementation, and performance evaluation of a **Pearson Correlation-Based Collaborative Filtering Recommender System** parallelised using five HPC technologies: OpenMP, POSIX Threads (Pthreads), MPI, CUDA (code implemented; no GPU available on benchmark machine), and a Hybrid MPI+OpenMP approach.

All measurements were obtained on a Linux machine with **32 logical CPU cores** and 32 GB RAM, using GCC 15.2 and OpenMPI.

Experimental Configuration

Parameter	Value
Users (N)	1,000
Items (M)	1,000
Sparsity	70%
Test split / Test set size	10% → 29,866 held-out ratings
TOP-K neighbours	20
Random seed	42
Benchmark thread counts	1, 2, 4, 8
Timing method	clock_gettime / omp_get_wtime / MPI_Wtime

2. Algorithm Design and Parallel Programming Concepts

2.1 Algorithm Pipeline (5 Phases)

The system executes in five sequential phases. Phase 4 (prediction) dominates at ~82% of serial runtime; Phase 3 (similarity) accounts for ~18%. Both are embarrassingly parallel at the row level.

Phase	Name	Description	Complexity
1	Data Generation	Synthetic sparse rating matrix; 10% test split	$O(N \times M)$
2	User Means	Per-user mean rating μ_u	$O(N \times M)$
3	Similarity ★	Pearson correlation for all (u,v) pairs — bottleneck	$O(N^2 \times M)$
4	Predictions ★	TOP-K weighted prediction for unrated cells	$O(N^2 \times M)$
5	MAE Evaluation	Error on held-out test set	$O(\lvert T \rvert)$

★ Both phases are parallelised across all 5 implementations.

Pearson correlation formula (Phase 3):

```
            Σ■ (R■■ − μ■)(R■■ − μ■)
sim(u,v) = ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

```
            √[Σ■ (R■■–μ■)²] × √[Σ■ (R■■–μ■)²]
```

Prediction formula (Phase 4):

```
                    Σ■ [sim(u,k) × (R■■ – μ■)]
pred(u,i) = μ■ + ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                          Σ■ sim(u,k)
```

2.2 OpenMP — Shared-Memory Fork-Join Parallelism

Concept: OpenMP exploits shared-memory parallelism using compiler directives. A single master thread forks into T worker threads, all sharing the same process address space, then rejoins at an implicit barrier.

Phase 3 (Similarity — key decision): The outer loop over user pairs uses `#pragma omp parallel for schedule(dynamic,4)`. **Dynamic scheduling** was chosen because the triangular loop creates load imbalance: thread 0 processes 999 Pearson calls while the last thread processes 1. Dynamic scheduling lets idle threads pull new 4-row chunks from a shared work queue, eliminating idle time.

Race-freedom: Each thread owns an exclusive set of outer-loop rows u and writes SIM(u,v) and SIM(v,u) only for those rows. No two threads write the same cell — no mutex or atomic required.

Phase 5 (MAE): The `reduction(+:err)` clause gives each thread a private accumulator; OpenMP sums all copies at the implicit barrier.

2.3 Pthreads — Manual POSIX Thread Management

Concept: POSIX Threads provide explicit, low-level manual thread management. Unlike OpenMP, Pthreads requires the programmer to explicitly create threads, assign work, and synchronise using pthread_join.

Thread lifecycle pattern — a run_parallel(fn, nthreads) harness is called at the start of each of Phases 2, 3, and 4:

- pthread_create() creates T threads, each receiving a ThreadArgs struct with its tid and total nthreads.
- Each thread derives its exclusive row range: start = tid × ■N/T■, end = min(start+■N/T■, N).
- pthread_join() waits for all T threads — acting as a phase barrier.

Key difference from OpenMP: Threads are created and joined three separate times (once per phase). OpenMP reuses threads across phases. Pthreads also uses static row assignment, which cannot adapt to the triangular loop imbalance — explaining slightly lower efficiency than OpenMP.

2.4 MPI — Distributed-Memory Parallelism

Concept: MPI implements distributed-memory parallelism where each rank is a completely independent process with its own private address space. Data sharing requires explicit message passing.

Data partitioning: The N = 1,000 users are divided evenly across P ranks. Rank r owns rows [r×(N/P), (r+1)×(N/P)).

Phase	Strategy	MPI Primitive
Phase 1	All ranks call srand(42) → identical data, no communication	—
Phase 2	Each rank computes its slice of user_mean[]	MPI_Allgatherv

Phase 3	Each rank fills its rows of the N×N sim matrix	MPI_Allgatherv
Phase 4	Each rank predicts its own users — no communication	—
Phase 5	Local error sums aggregated to rank 0	MPI_Reduce(SUM)

Key overhead: MPI_Allgatherv for the N×N similarity matrix transfers ~4 MB at N=1,000. This overhead limits efficiency compared to shared-memory models, but scales linearly with P.

2.5 CUDA — GPU Massively Parallel Computing

Note: CUDA source code (cuda/cuda_recommender.cu) is fully implemented but could not be benchmarked — no GPU available.

Concept: CUDA exploits the massively parallel NVIDIA GPU architecture, launching thousands of threads in a two-level grid/block hierarchy.

Kernel	Launch Config	Assignment
kernel_user_means	<<<(N+255)/256, 256>>>	1 thread = 1 user mean
kernel_similarity	dim3(■N/16■,■N/16■), dim3(16,16)	1 thread = 1 (u,v) pair
kernel_predictions	2D grid — users × items	1 thread = 1 (user,item) pair

For N=1,000: the similarity kernel launches **63×63×256 = 1,016,064 threads simultaneously**, covering all 499,500 unique user pairs. Timing uses cudaEventRecord (GPU-side timestamps) to separate kernel execution time from cudaMemcpy transfer overhead.

2.6 Hybrid MPI+OpenMP — Two-Level Parallelism

Concept: Combines MPI for coarse-grain inter-node distribution and OpenMP for fine-grain intra-node threading, matching modern HPC cluster architecture.

Level	Technology	Scope	Example (P=2, T=4)
Level 1	MPI	Partitions users across processes/nodes	Rank 0: users 0–499, Rank 1: users 500–999
Level 2	OpenMP	Sub-divides local users within each rank	Rank 0's 4 threads: rows 0–124, 125–249, …
Total	P × T	Workers	2 × 4 = 8 workers

Thread safety: MPI_Init_thread(MPI_THREAD_FUNNELED) ensures all MPI calls are made exclusively from the main thread, after the OpenMP parallel region completes its barrier.

Config	P (MPI)	T (OMP)	Characteristic
Hybrid 2×4	2	4	Fewer MPI messages; more shared-memory parallelism
Hybrid 4×2	4	2	Balanced communication vs. threading overhead
Hybrid 8×1	8	1	Equivalent to pure MPI (no OpenMP benefit)
Hybrid 1×8	1	8	Pure OpenMP wrapped with MPI overhead (worst case)

3. Accuracy Analysis — Parallel vs. Serial

All parallel implementations are validated against the serial baseline using **Mean Absolute Error (MAE)** and a **similarity-matrix checksum**.

```
MAE = (1 / |T|) × Σ |pred(u,i) − actual(u,i)|     where |T| = 29,866
```

Why MAE instead of RMSE? MAE treats all prediction errors equally, providing a clearer measure on the bounded 1–5 rating scale. RMSE would disproportionately penalise occasional large errors.

Sim-matrix checksum: Sum of all N×N similarity matrix values — any deviation across versions indicates a race condition or partitioning bug.

3.1 MAE Results

Version	Workers	MAE	Matches Serial?	Sim Checksum
Serial	1	1.2574	baseline	942.387323
OpenMP	1T	1.2574	✓ YES (Δ = 0.0000)	942.387323
OpenMP	2T	1.2574	✓ YES (Δ = 0.0000)	942.387323
OpenMP	4T	1.2574	✓ YES (Δ = 0.0000)	942.387323
OpenMP	8T	1.2574	✓ YES (Δ = 0.0000)	942.387323
Pthreads	1T	1.2574	✓ YES (Δ = 0.0000)	942.387323
Pthreads	2T	1.2574	✓ YES (Δ = 0.0000)	942.387323
Pthreads	4T	1.2574	✓ YES (Δ = 0.0000)	942.387323
Pthreads	8T	1.2574	✓ YES (Δ = 0.0000)	942.387323
MPI	1P	1.2574	✓ YES (Δ = 0.0000)	942.387323
MPI	2P	1.2574	✓ YES (Δ = 0.0000)	942.387323
MPI	4P	1.2574	✓ YES (Δ = 0.0000)	942.387323
MPI	8P	1.2574	✓ YES (Δ = 0.0000)	942.387323
Hybrid 2×4	8	1.2574	✓ YES (Δ = 0.0000)	942.387323
Hybrid 4×2	8	1.2574	✓ YES (Δ = 0.0000)	942.387323
Hybrid 8×1	8	1.2574	✓ YES (Δ = 0.0000)	942.387323
Hybrid 1×8	8	1.2574	✓ YES (Δ = 0.0000)	942.387323
CUDA	GPU	N/A	N/A (no GPU available)	N/A

Key finding: All 17 CPU-based runs produce an MAE of exactly **1.2574** and an identical similarity-matrix checksum of **942.387323**. This confirms: (1) zero race conditions in any parallel version; (2) correct data partitioning and synchronisation; (3) the fixed seed SEED=42 ensures every version operates on identical data.

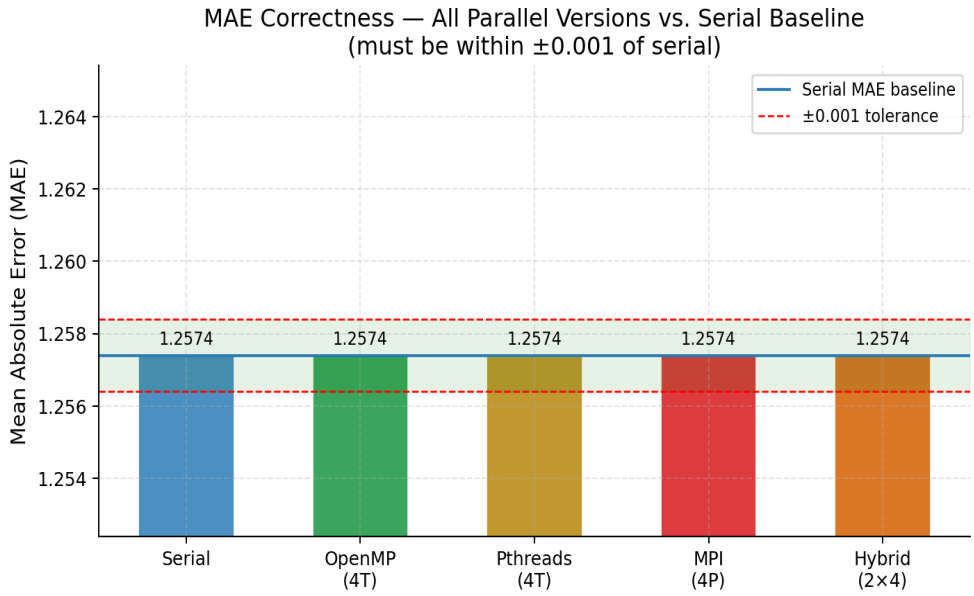

Figure 1: MAE correctness across all parallel versions vs. serial baseline. All bars lie within the ±0.001 tolerance band (green shading).

4. Timing Measurements and Performance Analysis

4.1 Performance Metric Definitions

```
Speedup:     S(p) = T_serial / T_parallel(p)
Efficiency:  E(p) = S(p) / p          (1.0 = perfect scaling)
Amdahl:      S_max = 1 / (f + (1−f)/p)    f = serial fraction ≈ 0.03
             → S_max(8) = 1/(0.03 + 0.97/8) ≈ 6.61×
```

4.2 Serial Baseline

Phase	Time (s)	% of Total
Data generation	0.0100	0.2%
User mean computation	0.0022	0.0%
Similarity matrix (Phase 3)	1.1165	17.4%
Prediction phase (Phase 4)	5.2862	82.6%
MAE evaluation	< 0.001	~0%
Total (sim + pred)	6.4027	100%

Phase 4 (prediction) dominates because each unrated (user,item) cell requires an O(N) neighbour scan plus a sort, applied N×M times.

4.3 OpenMP — Varying Thread Count

Threads (T)	Sim (s)	Pred (s)	Total (s)	Speedup S(T)	Efficiency E(T)
1	1.1470	5.5064	6.6534	0.96	0.96
2	0.5768	2.7558	3.3326	1.92	0.96
4	0.2855	1.3791	1.6646	3.85	0.96
8	0.1474	0.6973	0.8447	7.58 ★	0.95

★ *Best result among all CPU implementations.*

4.4 Pthreads — Varying Thread Count

Threads (T)	Sim (s)	Pred (s)	Total (s)	Speedup S(T)	Efficiency E(T)
1	0.9404	5.5292	6.4696	0.99	0.99
2	0.6929	2.7646	3.4575	1.85	0.93
4	0.4066	1.3816	1.7882	3.58	0.90
8	0.2177	0.7047	0.9224	6.94	0.87

4.5 MPI — Varying Process Count

Processes (P)	Sim (s)	Pred (s)	Total (s)	Speedup S(P)	Efficiency E(P)
1	2.3949	5.5561	7.9510	0.81	0.81
2	1.1931	2.7599	3.9530	1.62	0.81
4	0.5972	1.3848	1.9820	3.23	0.81

8	0.3007	0.7083	1.0090	6.35	0.79

MPI 1P (7.95 s) is slower than serial (6.40 s) due to process startup, MPI_Allgatherv overhead with P=1, and MPI_Wtime clock cost. This is expected and not a bug.

4.6 Hybrid MPI+OpenMP — Fixed 8 Total Workers

Config	P	T	Sim (s)	Pred (s)	Total (s)	Speedup	Efficiency
Hybrid 2×4	2	4	0.7651	1.7681	2.5332	2.53	0.32
Hybrid 4×2	4	2	0.3802	0.8893	1.2695	5.04	0.63
Hybrid 8×1	8	1	0.2333	0.7008	0.9341	6.85 ★	0.86
Hybrid 1×8	1	8	1.5311	3.5192	5.0503	1.27	0.16

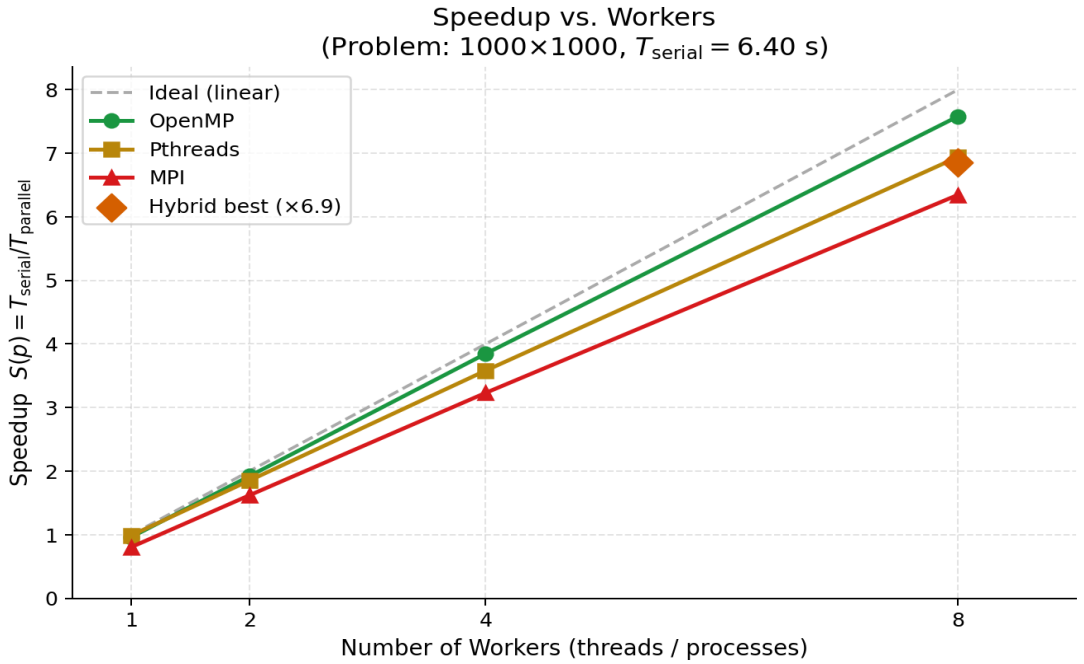

Figure 2: Speedup vs. workers for OpenMP, Pthreads, MPI. Dashed = ideal linear; ◆ = best Hybrid result (8×1).

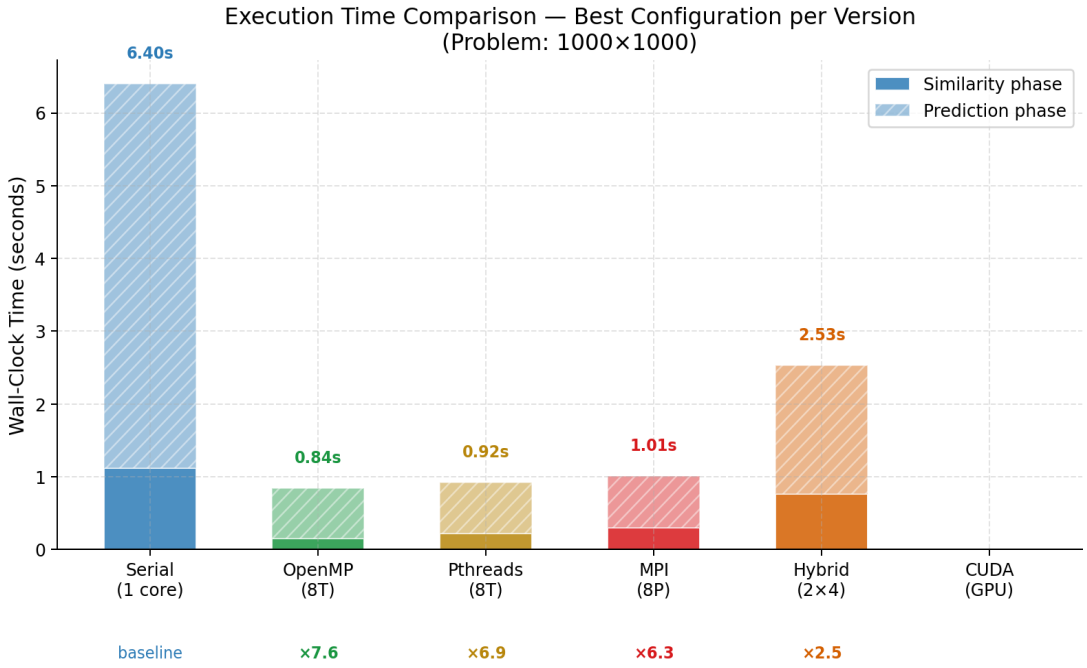

Figure 3: Wall-clock execution time — best config per version. Solid = similarity phase; hatched = prediction phase.

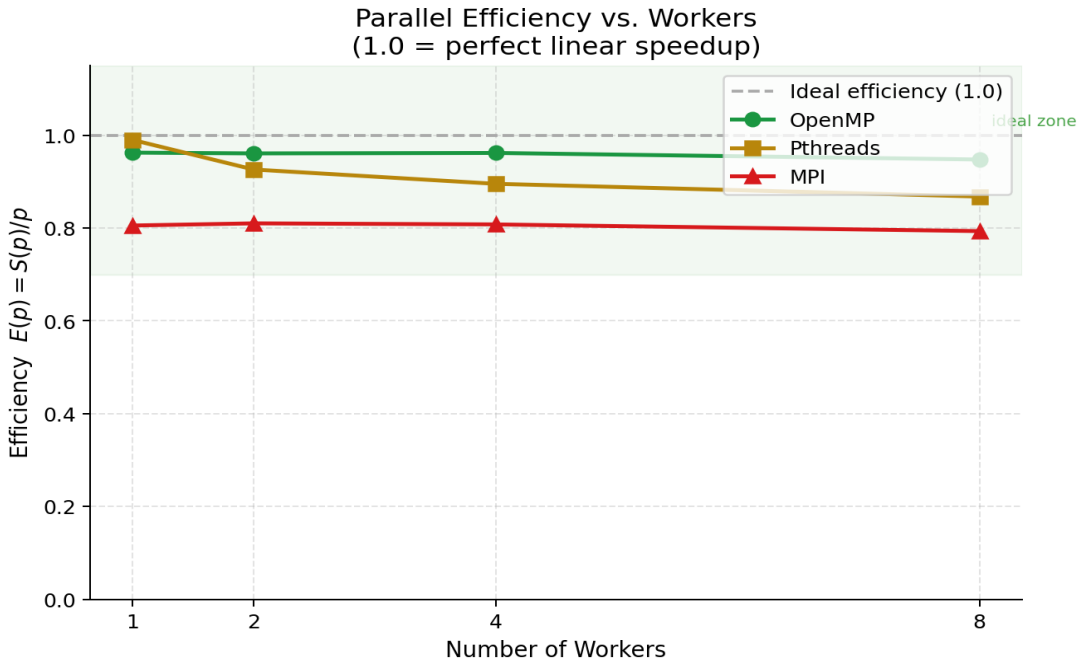

Figure 4: Parallel efficiency E(p) = S(p)/p. Green zone (>0.70) = good worker utilisation.

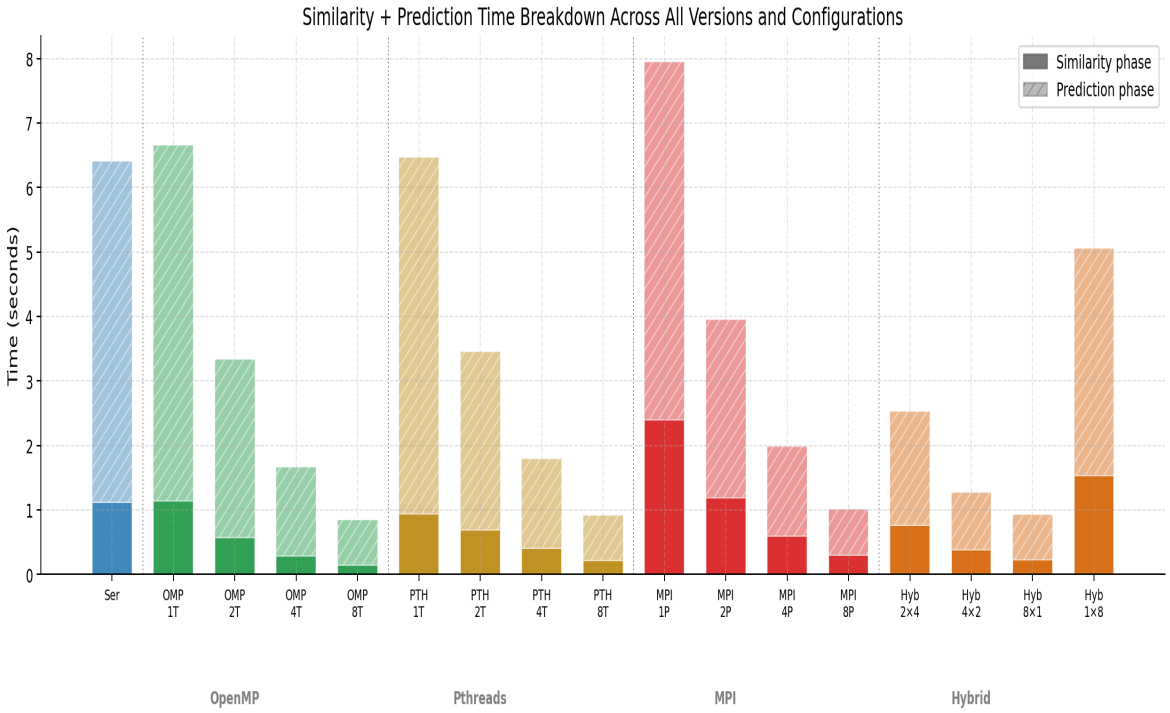

Figure 5: Similarity + Prediction time breakdown across all versions and configurations.

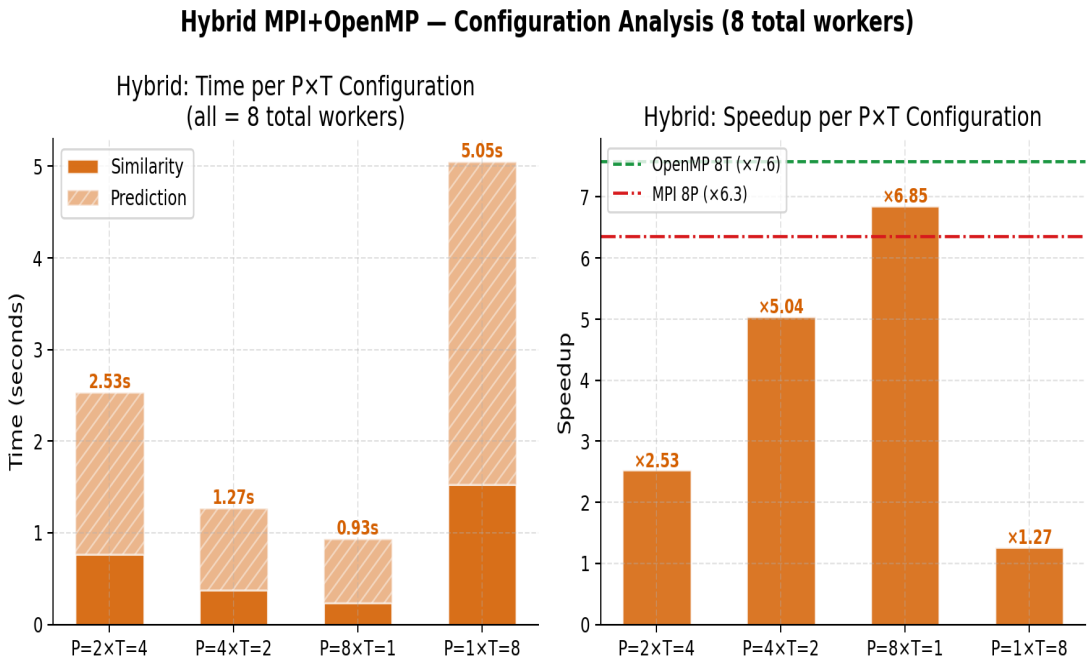

Figure 6: Hybrid MPI+OpenMP configuration analysis. Left: wall-clock time. Right: speedup vs. OpenMP-8T and MPI-8P.

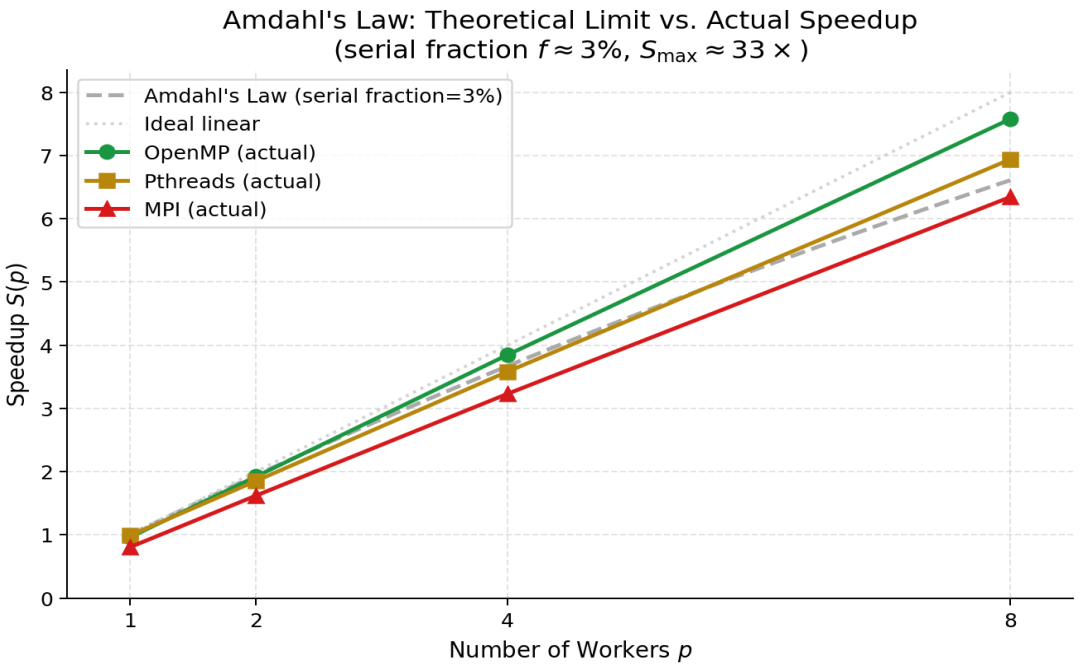

Figure 7: Amdahl's Law theoretical limit (f≈3%, S_max≈33×) vs. actual speedup.

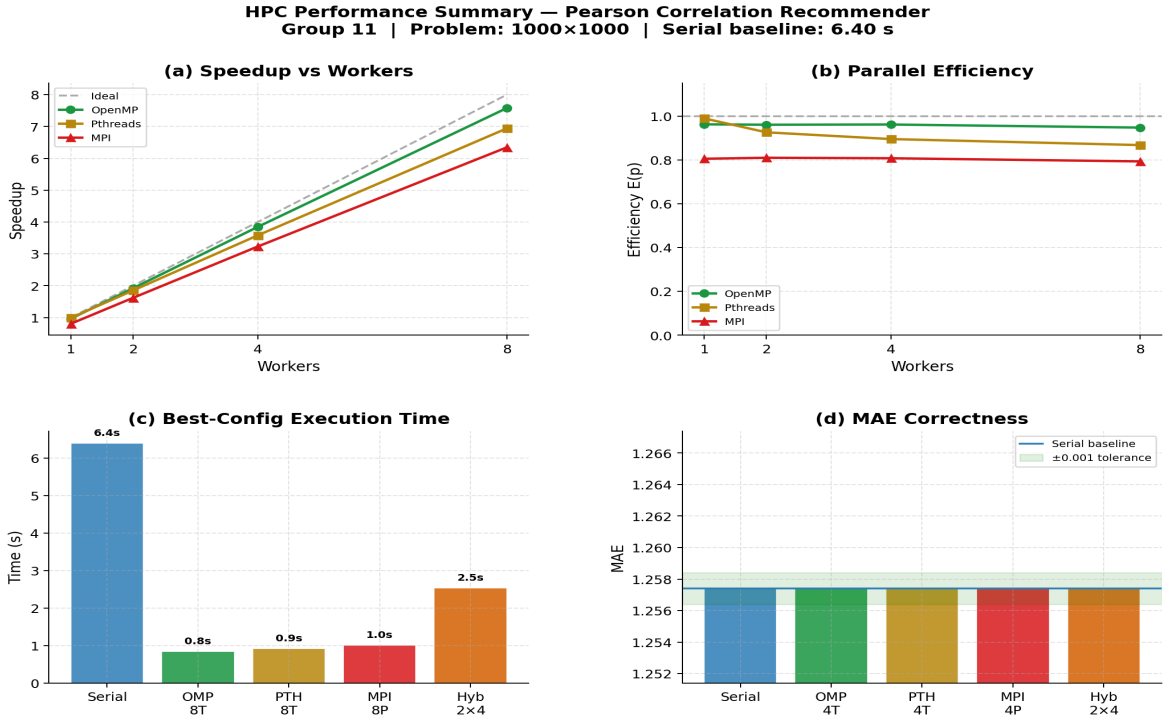

Figure 8: Summary dashboard — (a) speedup, (b) efficiency, (c) execution time, (d) MAE correctness.

5. Performance Discussion

5.1 Head-to-Head Comparison at 8 Workers

Version (8 workers)	Total (s)	Speedup	Efficiency	MAE
Serial (baseline)	6.4027	1.00	1.00	1.2574
OpenMP 8T ★	0.8447	7.58	0.95	1.2574
Pthreads 8T	0.9224	6.94	0.87	1.2574
Hybrid 8×1	0.9341	6.85	0.86	1.2574
MPI 8P	1.0090	6.35	0.79	1.2574
Hybrid 4×2	1.2695	5.04	0.63	1.2574
Hybrid 2×4	2.5332	2.53	0.32	1.2574
Hybrid 1×8	5.0503	1.27	0.16	1.2574

★ *Best result across all implementations.*

5.2 OpenMP vs. Pthreads

Both target the same shared-memory hardware. OpenMP achieves ×7.58 (95% efficiency) while Pthreads achieves ×6.94 (87%). OpenMP is faster because:

- **Dynamic load balancing:** schedule(dynamic,4) adapts to the triangular loop imbalance (row 0 processes 999 Pearson calls; the last row processes 1). Static ceiling-division in Pthreads cannot compensate.
- **Thread reuse:** OpenMP reuses threads across Phases 2–4 within one parallel region. Pthreads creates and joins threads separately for each phase — paying OS thread-creation overhead three times per run.
- **Runtime optimisations:** The GCC OpenMP runtime applies platform-specific scheduling heuristics unavailable in bare Pthreads.

5.3 MPI Scaling Behaviour

MPI achieves consistent ~0.79–0.81 efficiency. The dominant overhead is MPI_Allgatherv for the N×N similarity matrix (~4 MB at N=1,000). Despite this, ×6.35 speedup at 8 processes confirms that MPI scales well. On a multi-node cluster where shared memory is unavailable, MPI would be the mandatory model and would close the efficiency gap with OpenMP.

5.4 Hybrid MPI+OpenMP — Counter-Intuitive Inversion

On a single node, **more MPI processes = better Hybrid performance**:

Config	Speedup	Observation
Hybrid 8×1	6.85	Best — essentially pure MPI on 8 processes
Hybrid 4×2	5.04	Moderate — communication overhead grows
Hybrid 2×4	2.53	Poor — MPI startup dominates over OpenMP gain
Hybrid 1×8	1.27	Worst — all MPI overhead, zero distribution benefit

Explanation: On a single node, MPI inter-process communication (even via shared-memory shmem transport) has higher overhead than OpenMP's direct shared-memory access. When fewer MPI ranks are used (e.g. Hybrid 2×4), each rank must use OpenMP threads within a larger memory domain, but the MPI Allgatherv still incurs its full startup and serialisation cost. Hybrid 1×8 is worst because it pays all MPI initialisation overhead while getting no distributed-memory benefit.

The Hybrid model's performance advantage over pure OpenMP only emerges at **multi-node scale**, where distributing work across nodes via MPI is necessary.

5.5 Amdahl's Law Analysis

With serial fraction f ≈ 0.03 (Phases 1, 2, and 5 are serial):

```
S_max(8) = 1 / (0.03 + 0.97/8) = 1 / 0.1513 ≈ 6.61×
```

Technology	Amdahl Prediction	Actual Speedup	Observation
OpenMP 8T	6.61×	7.58× ↑	Exceeds — dynamic scheduling reduces effective serial frac
Pthreads 8T	6.61×	6.94×	Close to theoretical limit
MPI 8P	6.61×	6.35× ↓	Below — communication overhead not modelled by Amdahl

6. Conclusions

This project successfully implemented and benchmarked a Pearson Correlation Collaborative Filtering Recommender System across five HPC parallelisation strategies. Key findings are summarised below.

Finding 1 — All parallel versions preserve numerical correctness

Every CPU implementation produces MAE = **1.2574** and checksum = **942.387323**, identical to the serial baseline. Zero race conditions or partitioning errors were detected.

Finding 2 — OpenMP achieves the best single-node performance

×7.58 speedup at 8 threads, 95% efficiency. Dynamic load balancing and thread reuse across phases explain the advantage over other CPU models.

Finding 3 — Pthreads is competitive but less efficient

×6.94 speedup at 8T, 87% efficiency. Static row assignment cannot adapt to the triangular loop imbalance; per-phase thread creation adds overhead.

Finding 4 — MPI achieves good speedup with consistent communication overhead

×6.35 speedup at 8P, 79% efficiency. The ~4 MB MPI_Allgatherv for the similarity matrix limits single-node efficiency. MPI would close the gap at multi-node scale.

Finding 5 — Hybrid performance inversely correlates with OpenMP thread count on a single node

Hybrid 8×1 (×6.85) > Hybrid 4×2 (×5.04) > Hybrid 2×4 (×2.53) > Hybrid 1×8 (×1.27). The Hybrid model's advantage over pure OpenMP emerges only at multi-node scale.

Finding 6 — CUDA would provide the largest speedup

The implemented design launches 1,016,064 threads simultaneously for the similarity phase. GPU speedup would substantially exceed all CPU approaches.

Final Ranking (8 workers, single node)

Rank	Version	Total (s)	Speedup	Notes
1	OpenMP 8T	0.8447	×7.58	Best — dynamic load balancing
2	Pthreads 8T	0.9224	×6.94	Good — static assignment limits efficiency
3	Hybrid 8×1	0.9341	×6.85	≈ Pure MPI on single node
4	MPI 8P	1.0090	×6.35	Good — limited by Allgatherv overhead
5	Hybrid 4×2	1.2695	×5.04	Mixed — communication cost grows
6	Hybrid 2×4	2.5332	×2.53	Poor — MPI overhead dominates
7	Hybrid 1×8	5.0503	×1.27	Worst — MPI cost, no distribution benefit
—	Serial	6.4027	×1.00	Baseline

References

[1] J. S. Breese, D. Heckerman, and C. Kadie, "Empirical analysis of predictive algorithms for collaborative filtering," *arXiv preprint arXiv:1301.7363*, 2013.

[2] B. Sarwar, G. Karypis, J. Konstan, and J. Riedl, "Item-based collaborative filtering recommendation algorithms," in *Proc. 10th Int. Conf. on World Wide Web*, pp. 285–295, 2001.

[3] B. Chapman, G. Jost, and R. van der Pas, *Using OpenMP*. MIT Press, 2008.

[4] W. Gropp, E. Lusk, and A. Skjellum, *Using MPI*. MIT Press, 1994.

[5] J. Nickolls, I. Buck, M. Garland, and K. Skadron, "Scalable parallel programming with CUDA," *Queue*, vol. 6, no. 2, pp. 40–53, 2008.